Innovation & Entrepreneurship for Engineers

Dr Shankar

IIT Madras

2026

Innovation & Entrepreneurship for Engineers

- **Dr Shankar**

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  Mahindra & Mahindra
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- Ph.D (IISc)
- TEDx Speaker
- Prolific Inventor 40+ Patents



Innovation & Entrepreneurship for Engineers

Module 1
Engineers as Innovators

#1. What is Innovation ?
#2. Innovation through S & T

Module 2
Problem Discovery

#3. Finding Problems worth solving
#4. Customer Discovery & Design Thinking

Module 3
Creativity & Innovation

#5.Innovative Thinking – TRIZ, FLOW, CREATE
#6. Enhancing Innovation – AI - Collaboration

Module 4
Deep-Tech Entrepreneurship

#7. Deep – Tech Ventures
#8. Exponential & Systems Thinking

Module 5
Building a Startup

#9. Economics of Innovation & Entrepreneurship
#10. Business Models
#11. Intellectual Property

Module 6
Prototyping & Presentation

#12. Rapid Prototyping
#13. Pitch Your Idea
#14. Demo Day

Innovation Equation

Innovation = Idea + New Value + Path to Market

Innovation is the specific instrument of entrepreneurship.
It is the act that endows resources with a new capacity to create wealth – Peter Drucker

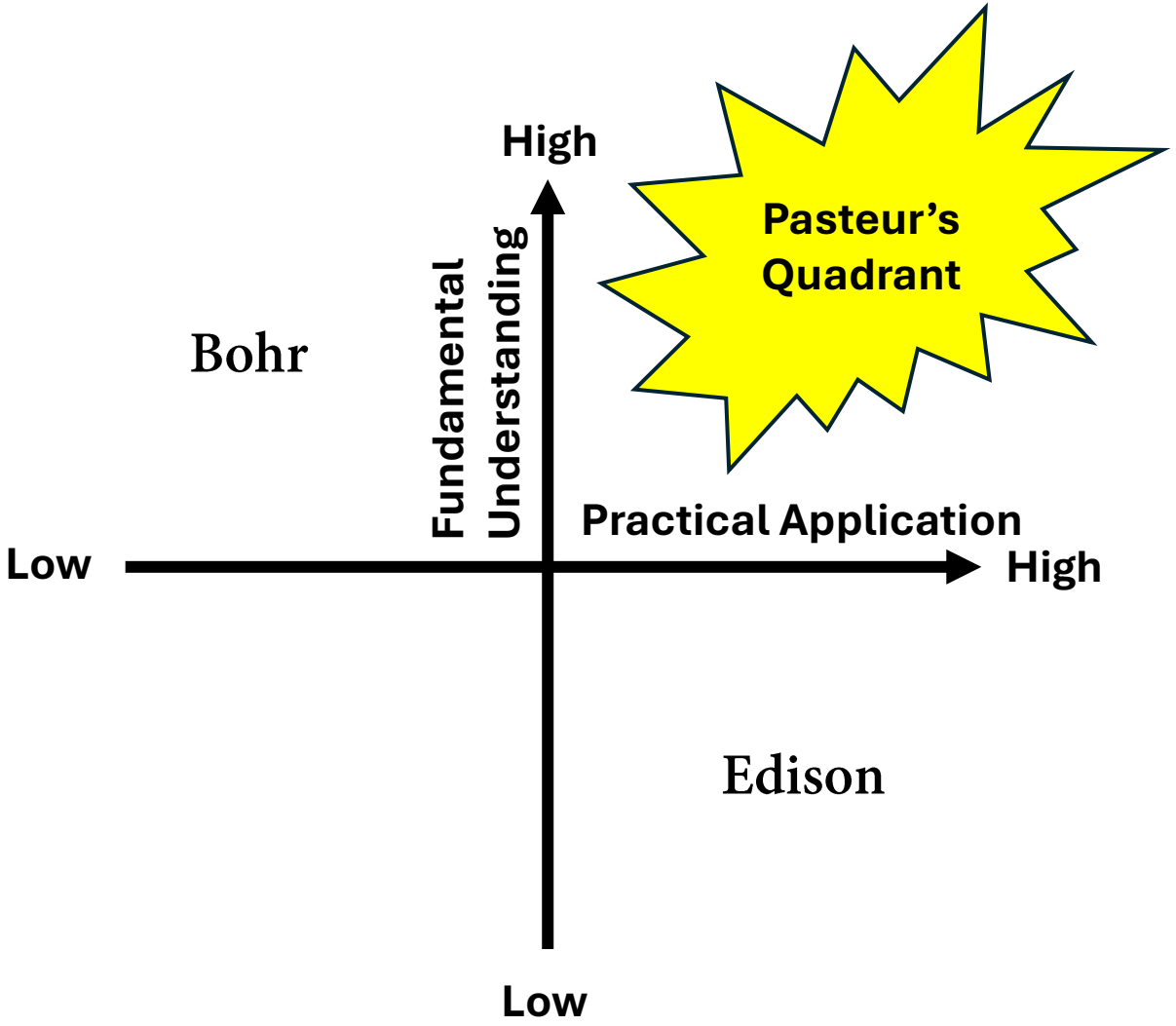
From Idea to Innovation

Idea → **Idea - Problem** → **Desirable Idea**
Scalable Idea
Sustainable Idea → **Feasible Idea**
Patentable Idea → **Viable Idea that creates value to the Customer**



Pasteur's Quadrant





Source -

Pasteur's Questions

- Why does this happen – Scientific Curiosity
- What fundamental mechanism is responsible – Deep Understanding
- Can that understanding solve an important problem ?
- Can the solution be translated into practice ?

Why does this happen – and can understanding it help us to solve an Important problem ?

Source -

Innovation & Entrepreneurship for Engineers

Finding problems worth solving

Dr Shankar

IIT Madras

28 August, 2026

Finding Problems worth solving

- Foundation – Key Concepts
- Practice – Examples and Case Study
- Practice - Exercise
- Summary
- Assignment



Sayantan is an innovation and entrepreneurship educator, consultant, and Founder of The Innovators' Garage.

Starting his career as an automobile engineer, he worked in R&D and corporate innovation at Mahindra and has since worked with organisations including Mahindra Automotive, CavinKare, UCO Bank, Mahindra Holidays, Ion Exchange, Lupin Pharmaceuticals and leading academic institutions.

His work focuses on helping people identify the right problems, uncover innovation opportunities, and convert them into practical action.

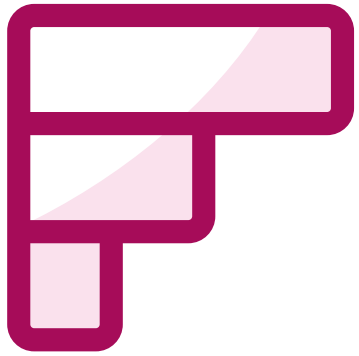
He teaches Design Thinking, TRIZ, opportunity identification, creativity and innovation strategy across corporate and academic audiences.

We want to find a worthy problem

We want to solve the problem innovatively

We want to scale it to an impactful venture

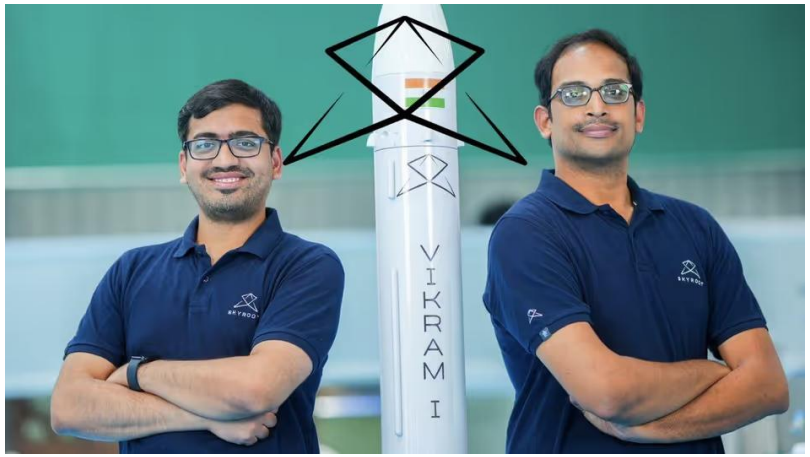
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Based on your strength, rank the three.
Rank 1 - I consider myself strong in this
Rank 2 - I consider myself fairly good in this
Rank 3 - I consider myself weak in this

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Pawan Kumar & Naga Bharath
SkyRoot Aerospace



Geetha Manjunath
NIRAMAI



Deepak Rajmohan
GreenPod Labs

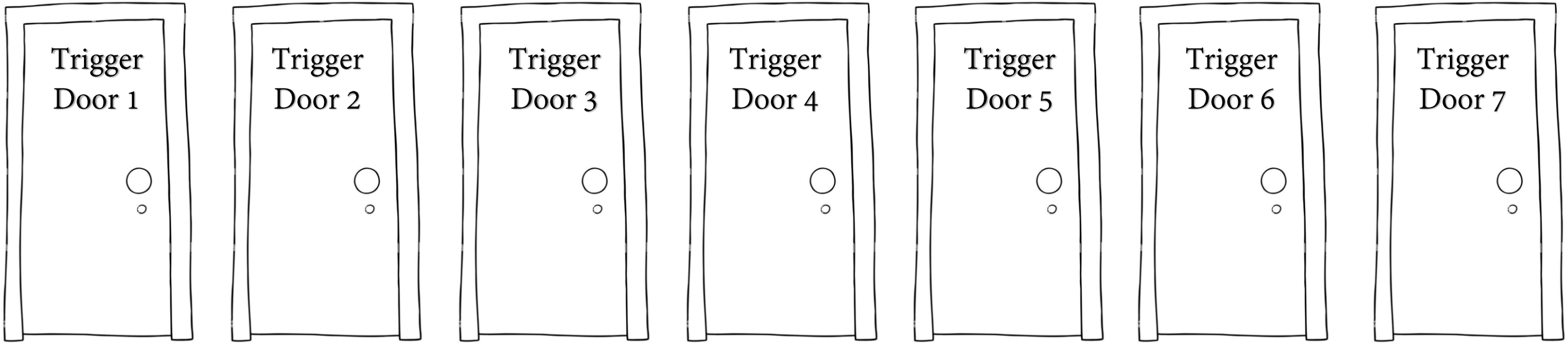
Is there a pattern?

Innovators start with finding an OPPORTUNITY

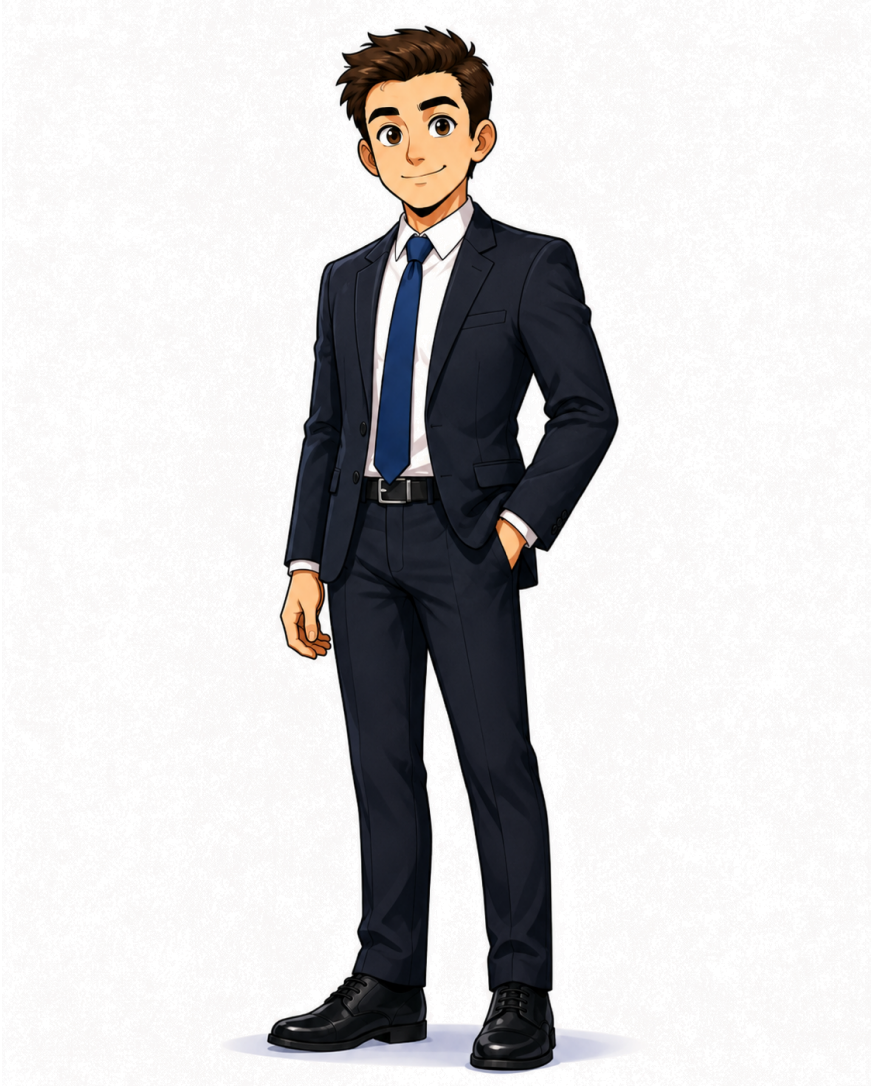
Then they check if that's worth exploring

And finally, they define the PROBLEM worth solving

Rather than waiting for an 'aha' moment, knock any of the Seven doors to help you find an opportunity



Meet Mr. Panini of modern days



Opportunity Trigger # 1



Leveraging unexpected Success – from Pharma to Software / Consulting for Pharma

Strides

Manufacturing, pharmacovigilance and regulatory-compliance problems

Pivot Path

Life. Science. Possibilities.

Exceptional internal success

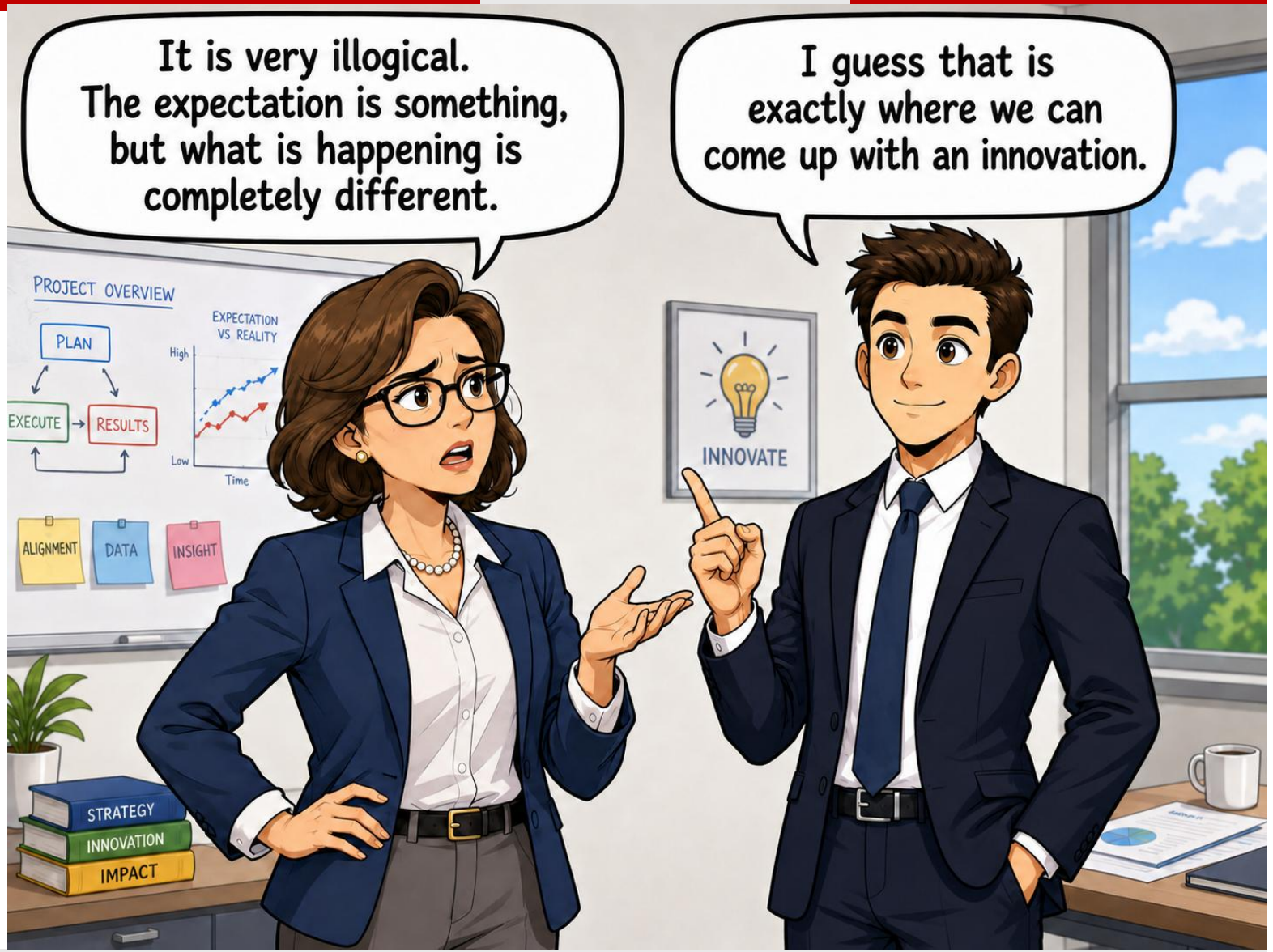
Today serves more than 70 pharma companies globally and in 2026 raised ₹100 crore at a valuation of roughly ₹230 crore

Opportunity Trigger 1

- Unexpected success or failure
- The bias is to stick to the intentional outcome
- Opportunity is to look at that without bias

Opportunity

Trigger # 2



Opportunity Trigger 2

People vote to **democratically** choose their representative

But most of the times, people **do not know a thing** about who they are voting

They just vote for a **symbol**

The **expectation** is people should **vote with knowledge**, but in **reality**, they **have zero knowledge** about the person they are voting for


the better india
This 16-YO Tech Whiz Built An AI Tool That Guided 70000 Voters In Tamil Nadu
tellmystory

Opportunity Trigger 2

- Something does not add up
- A difference in expectation and reality or results
- Opportunity is to investigate what's the odd thing out and do something about it

Opportunity

Trigger # 3



Opportunity Trigger 3

THIS STORY IS FROM JULY 22, 2023

WhatsApp pothole pics, KMC mobile vans will carry out repairs in 24 hours

But it hardly worked

Most people showed frustration..
But only a few asks, "Why isn't it working?"



Opportunity Trigger 3

- Something inside a process is weak / missing

- Opportunity is to add something good to it or

  remove the weak link and make the process better

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What is the major difference between Trigger 2 and Trigger 3

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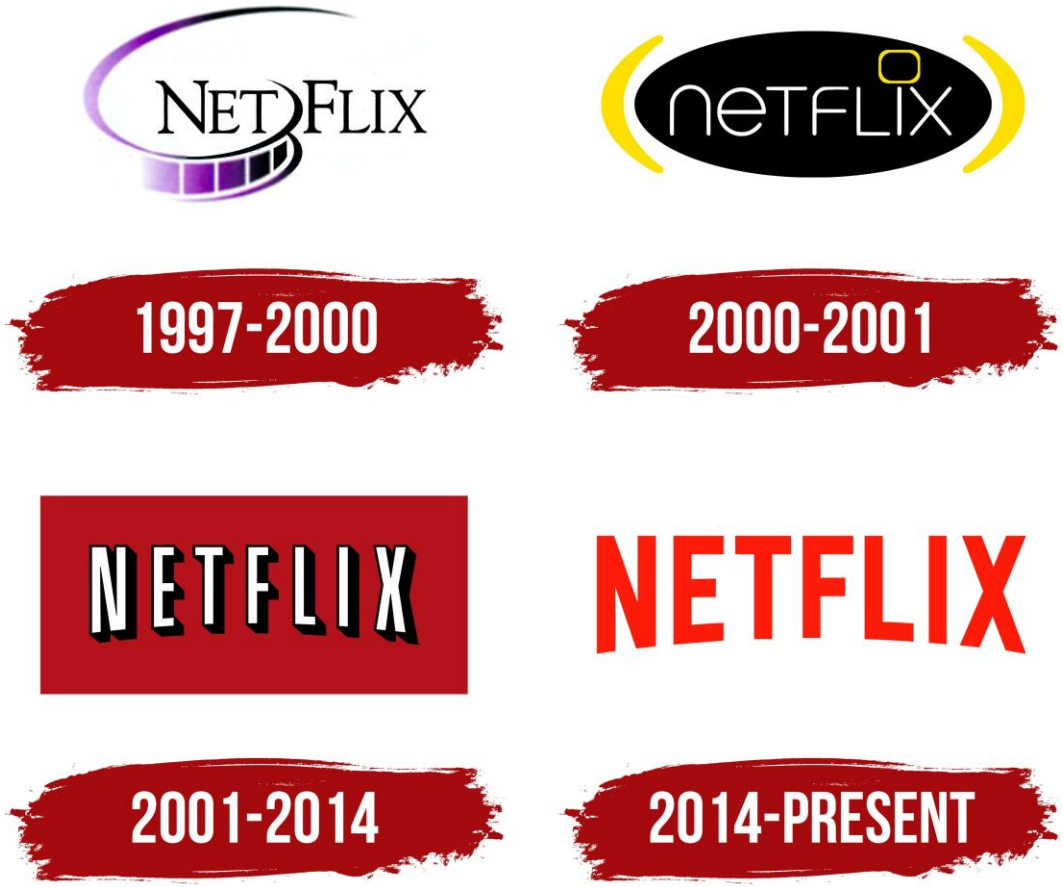
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Opportunity

Trigger # 4



Opportunity Trigger 4



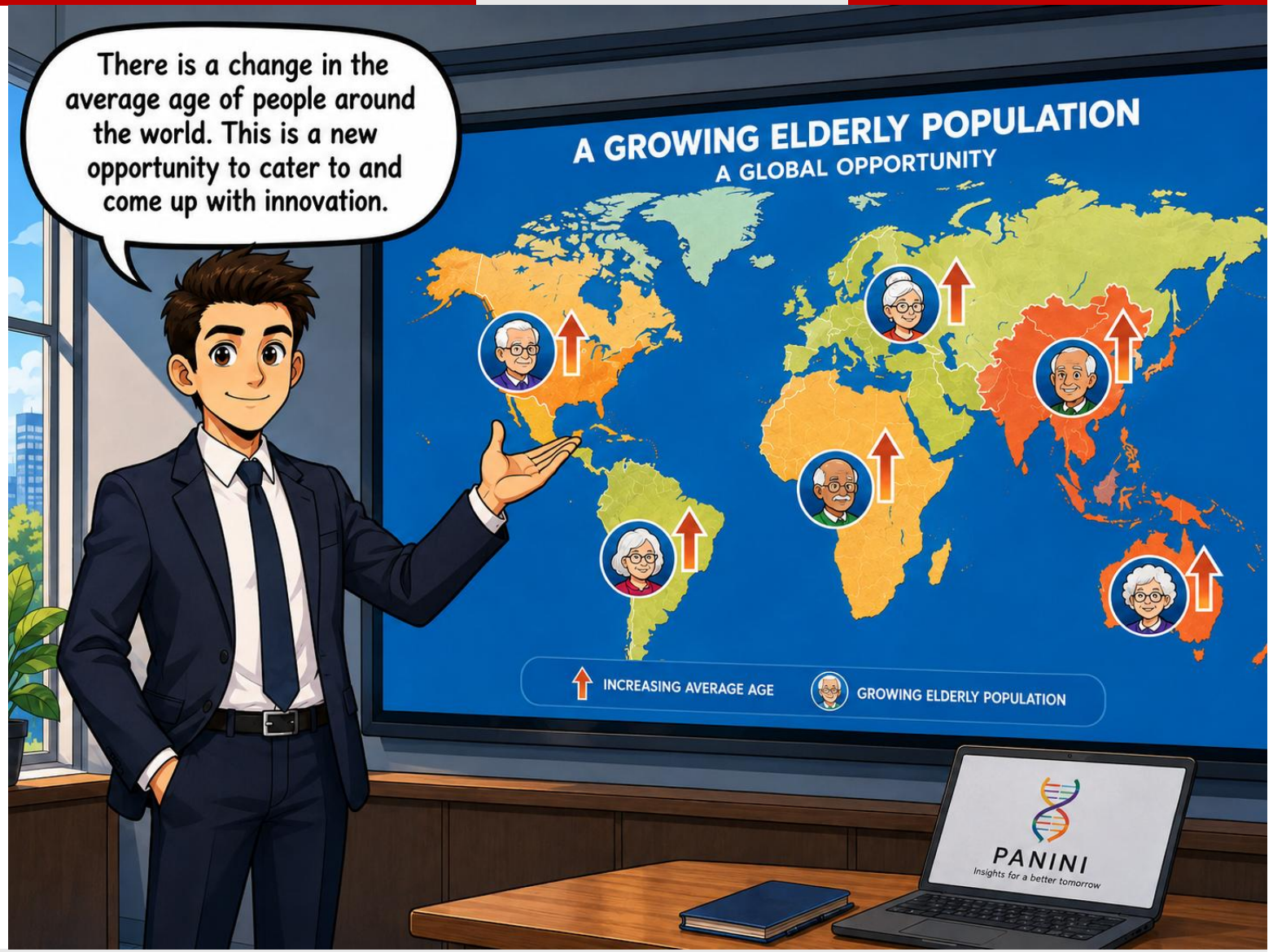


Opportunity Trigger 4

- A certain industry need, expectation or technology is changing fast
- The bias is to still hold on to the past
- Opportunity is to recognise the upcoming change faster and innovate around it

Opportunity

Trigger # 5



Opportunity

Trigger # 5

Real estate brands mostly focus on the young to mid level population to design their projects but...

Then there is

ashiana®
you are in safe hands



Opportunity Trigger 5

- The characteristics of people are changing – age, sex, migration pattern
- Opportunity is to predict what need/ situation this change would create in the long term
- Innovations based on this type of opportunities are most reliable because the ‘predicted need’ is backed by data

Opportunity

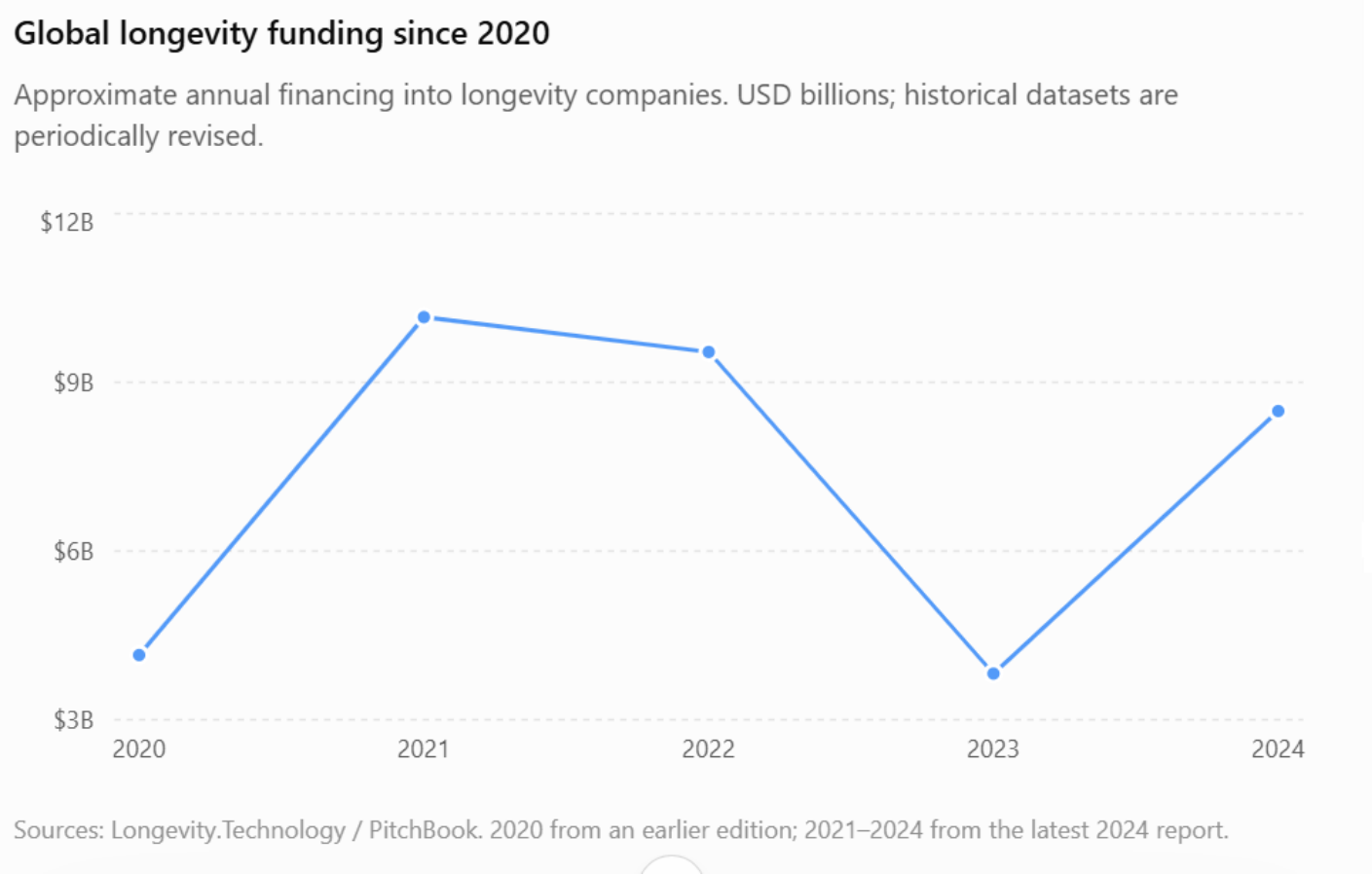
Trigger # 6



https://factfulnessquiz.com/

Opportunity Trigger 6

Gapminder.org



The average length of life worldwide is roughly 72, today. What was the global life expectancy a hundred years ago?

Answer options

A: 37 years (Correct)

B: 47 years (Wrong)

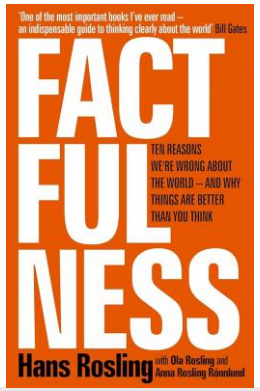
C: 57 years (Very wrong)

Survey Results

Of the people we have tested, **75%** got this question wrong.

Respondents





Opportunity Trigger 6

- Sometimes what people believe and what is the reality might be very different
- And sometimes the same thing can be perceived differently
- The bias is to try and educate people on one version
- But opportunity is to use the perception to come up with innovations

Opportunity

Trigger # 7



Opportunity Trigger 7

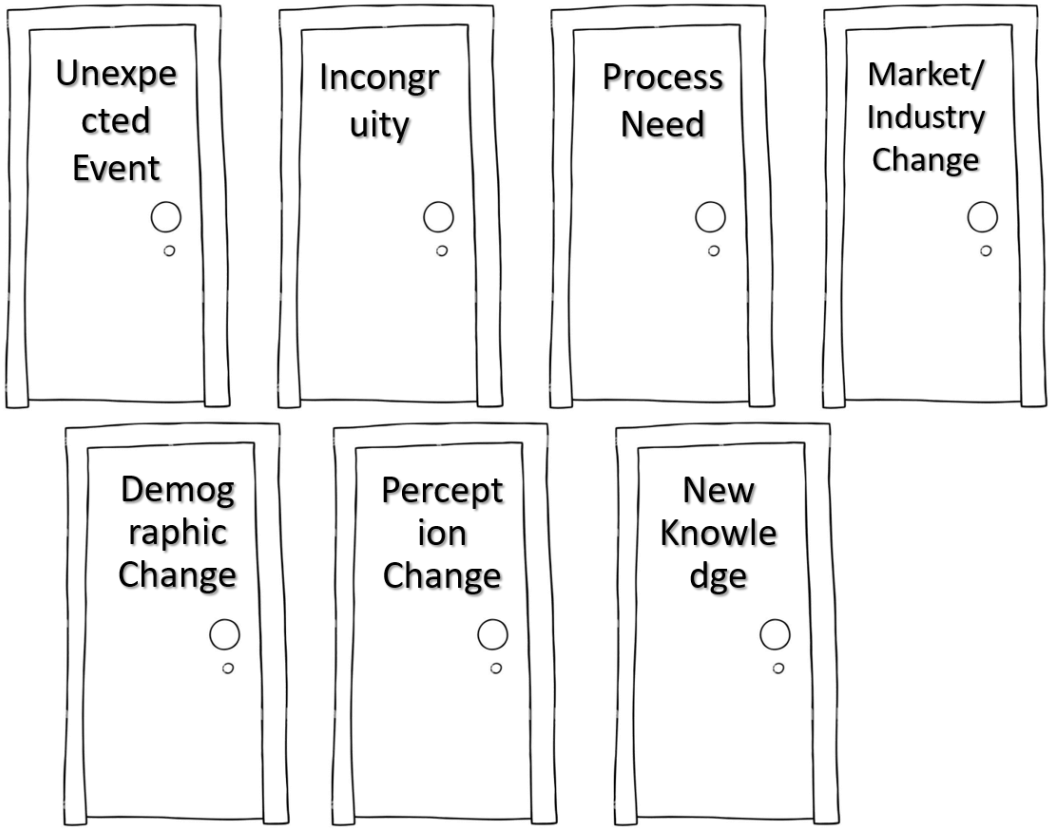


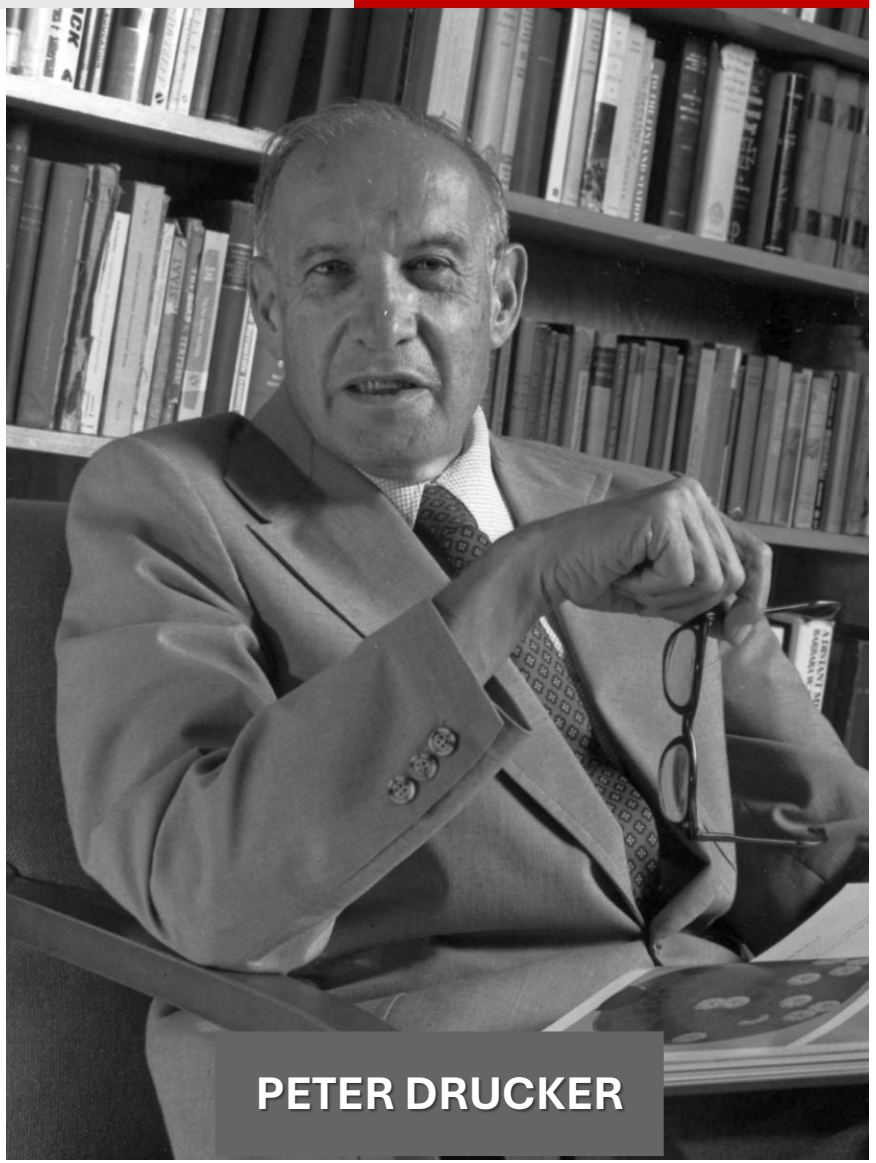
Opportunity Trigger 7

- Sometimes a new knowledge becomes available
- And sometimes an existing knowledge opens up new insights to a curious mind
- The opportunity is to identify the features of the new insights and find applicable benefits in the market

Each of these ‘Trigger Doors’ got a name

The seven sources of innovation





Finding Problems – Foundational Concepts

Internal:

- The Unexpected
- The Incongruous
- Process Need
- Industry & Market Structure Changes

External:

Demographics

Changes in Perception

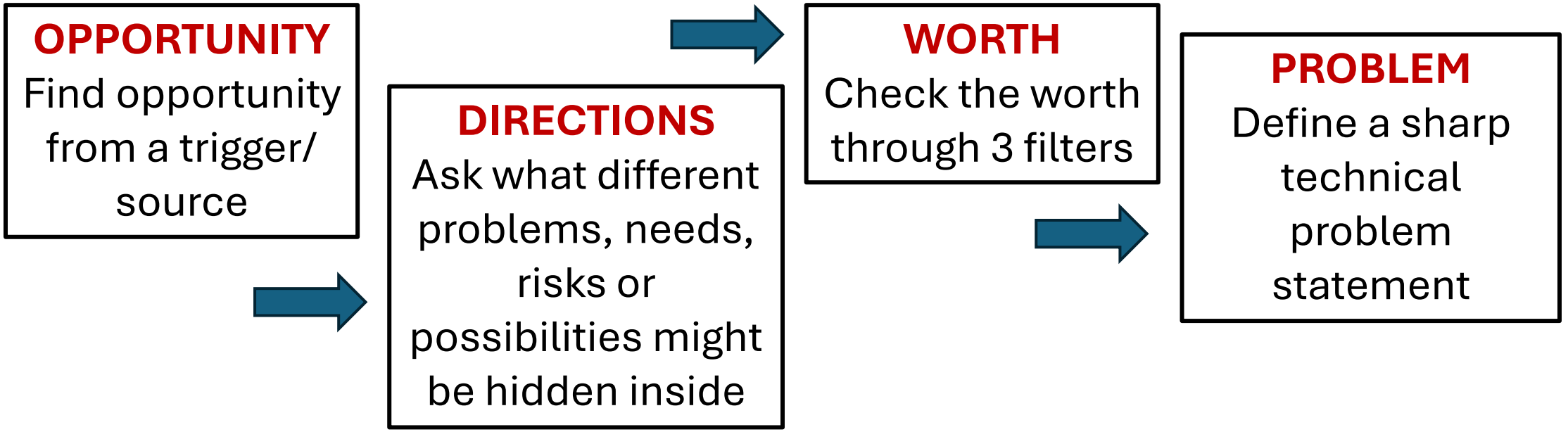
New Knowledge

7 Sources of Innovation – Peter Drucker

But for innovators an opportunity is not enough

We need problem statements to begin our Innovation Journey

How to define a PROBLEM STATEMENT from an OPPORTUNITY



Trigger – Question Mapping

Sources of Innovation	One simple question to ask
Unexpected Event	What new thing does this surprise make us look at?
Incongruity	What is not matching here?
Process Need	Which part of this process needs to work better?
Industry / Market Change	Because the market is changing, what will people or businesses need differently?
Demographic Change	Because people are changing, what will they need differently?
Change in Perception	Because people now think differently, what will they want differently?
New Knowledge	Where can this new ability be useful?

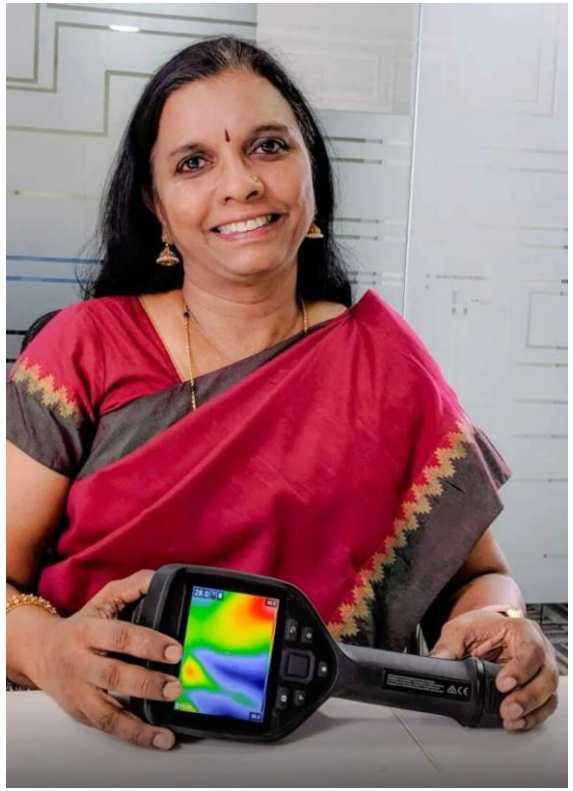
We get qualitative direction statements when we ask these questions

Then we check the worth of each

Checking which direction statement is worth pursuing – three filter check

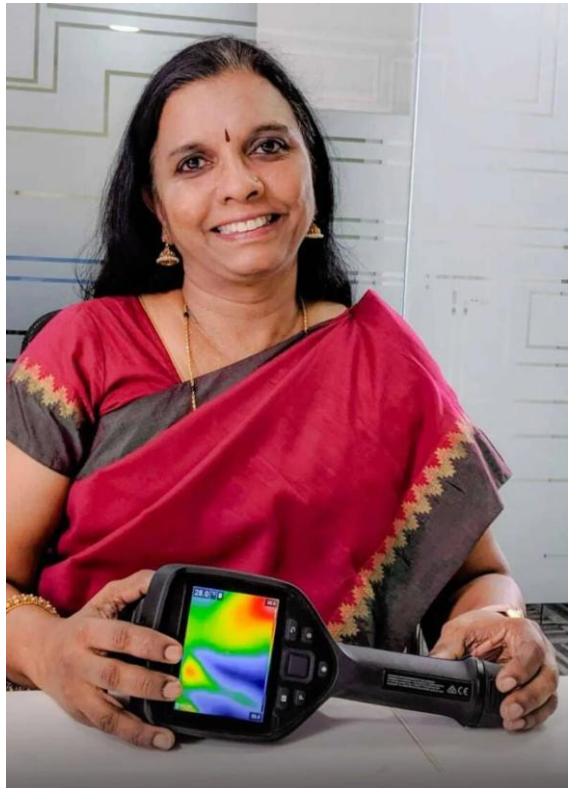


Example:



How Ms. Geetha Manjunath might have followed the process to come up with NIRAMAI?

Example:



NIRAMAI founder Geetha Manjunath

The observation used as opportunity was…

Breast-cancer screening is not reaching enough women early and regularly

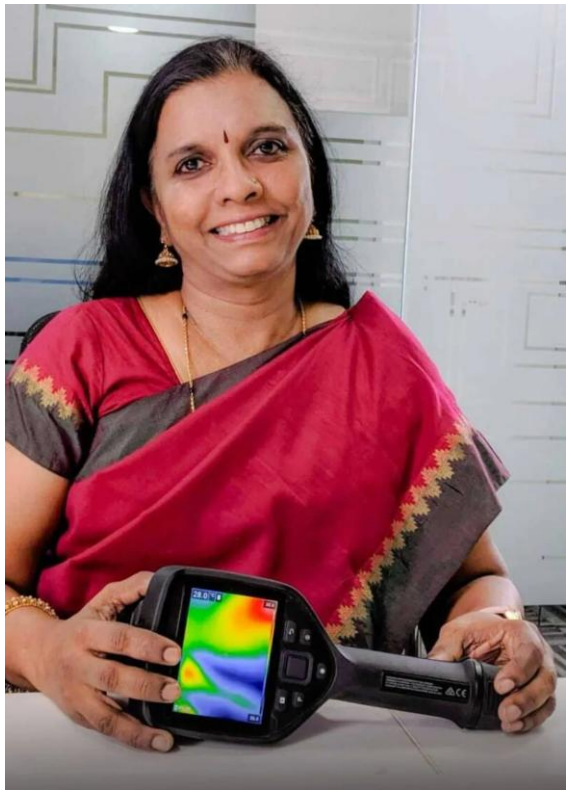
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Which Peter Drucker's Source of Innovation, Ms Geetha might have used as the trigger?

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NIRAMAI founder Geetha Manjunath

Example: Process Need

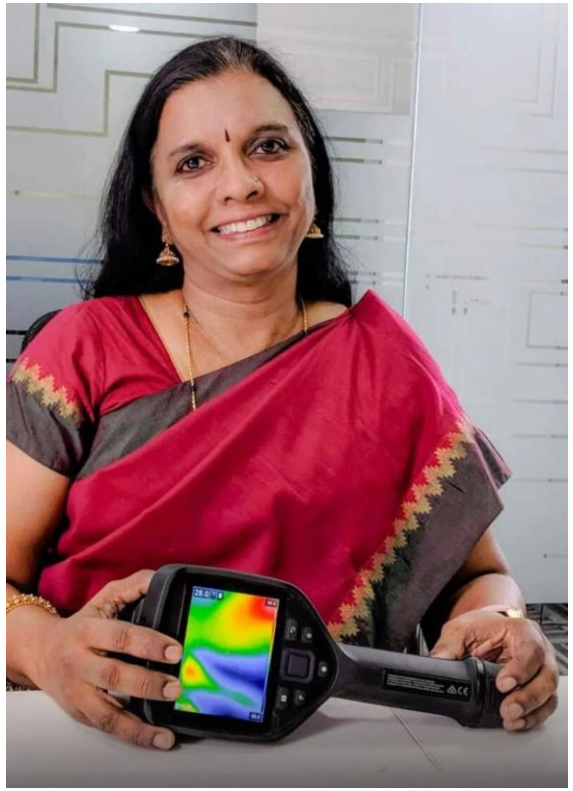
Breast-cancer screening is not reaching enough women early and regularly

She asked - Which part of this process needs to work better?

Qualitative Direction Statements	What it means
Awareness	Women may not know when or why they should get screened.
Access	Screening centres may be too far away or concentrated in hospitals.
Comfort	The existing screening experience may discourage some women.
Equipment	Screening may depend on large or specialised equipment.
Specialists	The process may depend too heavily on trained radiologists or technicians.

Direction	Mass	Money	Consequence	Worth
Awareness	Very High	Medium	Very High	High
Access	Very High	High	Very High	**Very High**
Comfort	High	High	High	High
Equipment dependency	Very High	Very High	Very High	**Very High**
Specialist dependency	High	High	High	High

Example:



NIRAMAI founder Geetha Manjunath

Final Statement based on the worth check

Women who need preventive breast-cancer screening have limited access because existing screening depends on specialised hospital-based equipment

Technically defined Problem Statement

How might we make breast-cancer screening accessible to women without depending on specialised hospital-based equipment?

Questions to check worth of opportunity direction

M Mass — How big is the affected group?

Check whether this direction affects enough people, customers, organisations, or situations.

Three check questions

1. How many people, customers, or organisations face this direction?
2. How often do they face it: rarely, monthly, weekly, daily, or continuously?
3. Is the affected group growing, stable, or shrinking?

Four-level scoring

LOW	Very few people; rare occurrence; limited growth.
MID	Noticeable group; occasional or periodic occurrence.
HIGH	Large group; recurring issue; stable or growing.
VERY HIGH	Very large or fast-growing group; frequent issue.

Rule: Give a score only after writing one estimate, number, example, or evidence point.

M Money — Is there enough economic value?

Check whether someone has a strong financial reason to solve this direction.

Three check questions

1. How much money is being spent, lost, or wasted because of this?
2. Who would pay to solve it, and how much could they reasonably pay?
3. Are people already paying for alternatives, workarounds, or partial solutions?

Four-level scoring

LOW	No clear payer; little visible spending or loss.
MID	Some payer interest; small or uncertain value.
HIGH	Clear payer; meaningful current spend or loss.
VERY HIGH	Strong payer; large spend/loss; urgent economic value.

Rule: Give a score only after writing one estimate, number, example, or evidence point.

C Consequence — What happens if nobody acts?

Check the seriousness of leaving this direction unsolved.

Three check questions

1. What is the measurable cost of doing nothing? Also consider opportunity cost
2. How serious can the worst reasonable outcome become?
3. How quickly does the consequence become serious?

Four-level scoring

LOW	Mostly inconvenience; easy to ignore for long.
MID	Meaningful loss, delay, or disruption.
HIGH	Serious financial, operational, health, or safety impact.
VERY HIGH	Severe harm, major loss, systemic risk, or national impact.

Rule: Give a score only after writing one estimate, number, example, or evidence point.

Finding problems worth solving – Assignment M2S3B1

- Go around the campus and scan the seven opportunities and select one of the opportunities.
- Translate the opportunity into qualitative direction statements by using the core question for the particular trigger / source of innovation (Charts #47)
- Now use the set of questions of the Mass Money Consequence checklist and quantify the problem statement (Charts #57 - 59)
- Look at different directions to solve the problem and create a comparison table (chart #54)
- Formulate a crisp technical problem statement
- Shoot a short video describing the complete process you followed
- You will be scored for (a) following the systematic process and (b) the quality of the problem statement.

(You and your team will work on this problem till the end of this course - generate ideas, build proto and test it - you will present the idea on the Demo day in October).

Format – Short video– Assignment due on 5th September 5 pm

Summary of Key Ideas

- Innovators don't just see; they observe through the Innovator's lens. They even look at surprising / difficult / illogical situations as innovation opportunities !
- Peter Drucker's Seven Sources of Innovation serve as triggers to help identify where such opportunities exist
- Each opportunity is analysed through sharp questions to generate multiple possible directions for innovation.
- Do not jump at the first direction. Different answers may reveal very different problems worth solving. Carefully analyze and compare all the directions and then focus on the most promising directions.

Summary of Key Ideas

- Use Mass, Money and Consequence to test worth.
  Mass: How many are affected?
  Money: Is there enough economic value to solve it?
  Consequence: What is the cost of not solving it?
- The strongest opportunities emerge strongly when scored for all these three filters.
- The final step is to convert the selected high-worth direction into a sharp technical problem statement that is specific enough to start your Innovation Journey.

Finding problems worth solving – References

https://hbr.org/2002/08/the-discipline-of-innovation

- https://www.youtube.com/watch?v=w_1Cow_iXTY
- https://www.ted.com/talks/hans_and_ola_rosling_how_not_to_be_ignorant_about_the_world
- https://www.ted.com/talks/hans_rosling_the_good_news_of_the_decade_we_re_winning_the_war_against_child_mortality
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